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A Multi-Axis Framework for Evaluating the Faithfulness, Stability, and Fairness of SHAP and LIME Explanations in Mental-Health Treatment Prediction

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***Abstract*—Post-hoc explanation methods such as SHAP and LIME are increasingly used to justify machine-learning predictions in mental-health screening tools. However, their reliability along axes beyond raw predictive accuracy is rarely audited end to end. This paper presents a leakage-controlled, reproducible evaluation pipeline that trains four classifiers (XGBoost, LightGBM, Random Forest, and Logistic Regression) on a large public mental-health survey with 77,708 records to predict treatment-seeking behavior. The resulting SHAP and LIME explanations are subjected to a four-axis audit: predictive significance, faithfulness (deletion/insertion area over the perturbation curve, AOPC), stability (empirical Lipschitz continuity), and fairness (outcome-level parity gaps and explanation-level demographic parity gaps, EDPG). Although XGBoost achieved the best discrimination (AUC = 0.876, 5-fold CV) and the highest faithfulness score (0.206), a Random Forest model provided significantly more Lipschitz-stable SHAP attributions (mean L = 0.051 vs. 0.39–0.42 for boosted trees). At the same time, SHAP and LIME achieved only moderate agreement (mean per-instance Spearman = 0.49) and found large gender disparities in both outcome parity (approximately equal-opportunity gap = 0.434) and explanation parity (after Bonferroni correction, 14 out of 15 features were significant across sub-segments). This paper concludes with considerations on the deployment of explainable mental-health screening tools, proposing a full leak-free pipeline.**

***Index Terms*—Explainable Artificial Intelligence, Algorithmic Fairness, Feature Attribution, Model Auditing, Mental Health Informatics.**

I. Introduction

The increased need and low cost of screening through digital formats have motivated the rapid growth of machine learning (ML) research in mental-health-risk and treatment-seeking-behavior detection from survey and social-media data [1]. As these predictions can affect referral, resource allocation or an individual's perception of their own risk post-hoc explanation methods, predominantly SHapley Additive exPlanations (SHAP) and Local Interpretable Model-agnostic Explanations (LIME), are commonly applied to such models in an effort to explain model outputs to clinicians. However, an explanation is only helpful if it correctly represents the reasoning of the model, is robust to small perturbations in input, and equally reliable across demographic subgroups. Because these properties are seldom verified prior to deployment, Rudin [2] has argued that post-hoc explanations of black-box models represent a real danger for high-stakes decisions unless they have been audited.

Most applied work in this area either reports SHAP summary plots and LIME examples and quits there without quantifying the extent to which the explanation actually approximates the model decision function, whether it holds with minor resampling, or whether the explanation is equally informative for men and women. This gap is particularly consequential in mental health, where model errors and unreliable explanations are not equally distributed across the population and can compound existing disparities in access to care [3].

This paper makes the following contributions:

1) A leakage-controlled, fully reproducible pipeline for predicting treatment-seeking behavior on a large ($N = 77,708$) public mental-health survey dataset [4], comparing four classifiers with proper cross-validated significance testing.
2) A four-axis evaluation framework for SHAP and LIME explanations covering faithfulness, stability, cross-method agreement, and fairness.
3) An explanation-level fairness audit (EDPG) that is, to our knowledge, rarely reported jointly with outcome-level fairness metrics in the mental-health ML literature.

II. Related Work

SHAP [5] generalizes many additive feature-attribution methods under a single framework based on game theory (the Shapley value). LIME [6] fits a local, sparse surrogate model around each prediction by simply sampling perturbations in the vicinity. Although SHAP and LIME address different questions with global Shapley decomposition and local linear

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approximation, both are widely adopted. However, recent work shows neither is unconditionally reliable; Huang and Marques-Silva [7] catalogued formal failure modes of Shapley-value attributions, and Hwang et al. [8] demonstrate that SHAP rankings are highly sensitive to arbitrary choices in feature representation. Those two work, together, motivate treating agreement between SHAP and LIME as an empirical question rather than an assumption.

Faithfulness is often implemented with deletion/insertion procedures where features are added or suppressed in order of attribution magnitude and the corresponding model output change is measured. Recent works formalizing this approach include ERASER benchmark [9] for tabular scenarios. Stability, on the other hand, questions if similar inputs receive similar explanations. Alvarez-Melis and Jaakkola [10] formalize this as an empirical Lipschitz continuity estimate, while Yeh et al. [11] show that faithfulness (which they term infidelity) and stability (sensitivity) can trade off against one another depending on model complexity. Karagoz et al. [12] benchmarked local and global XAI methods on tabular healthcare data using a similarly broad set of criteria, but, like most benchmarking work in this space, evaluated faithfulness and stability without a parallel fairness audit of the explanations themselves. Hassan et al. [13] combined four resampling strategies (ROS, RUS, SMOTE, ADASYN) with a stacking ensemble on a smaller ($N = 334$) mental-health survey and reported a 5-fold cross-validated accuracy of 99.9%. The near-perfect accuracy on such a small, heavily resampled sample is itself suggestive of the leakage risk that our fold-internal resampling protocol (Section III) is designed to avoid.

Group fairness metrics such as demographic parity and equalized opportunity [14] audit a model's outcomes across protected groups but provide nothing about whether its explanations are equally reliable for those groups. Balagopalan et al. [15] showed empirically that explanation fidelity could differ systematically by subgroup even when outcome-level fairness gaps were modest. Hwang et al. [8] further show this sensitivity can itself be group-dependent, motivating our inclusion of an explanation-level demographic parity gap (EDPG) alongside conventional outcome, fairness metrics. In mental health specifically, Park et al. [3] audited gender bias in a mobile-phone-based mental-health assessment model and established that predictive gender disparities are a known risk in this domain. However, study [3], like most applied mental-health ML work, evaluated fairness only at the outcome level and did not examine whether the model's explanations were themselves biased.

On the predictive side, several recent studies apply gradient-boosted trees with SHAP and LIME to mental-health prediction. Ahmed et al. [16] trained XGBoost, Random Forest, Logistic Regression, and a neural network on wearable actigraphy data to detect and stage depression, reporting 84.94% accuracy with XGBoost and using Holm-adjusted pairwise significance testing to confirm its advantage. Closer to our data source, Retnaswamy et al. [17] trained Logistic Regression, k-nearest neighbors, Random Forest, and a Decision Tree on the same datset [4] used in this study, reporting a best accuracy of 81.2% (Random Forest, AUC = 0.813) but neither cross-validating with paired significance tests nor evaluating any explanation beyond feature importance rankings. Chancellor and De Choudhury [1] noted at a review level, that predictive mental-health ML work rarely validates its explanations rigorously or audits them for subgroup fairness. Our study addresses this composite gap directly through reporting cross-validated statistical model comparison, deletion/insertion faithfulness, Lipschitz stability, SHAP–LIME agreement, and dual outcome-/explanation-level fairness audits within a single leak-free pipeline.

III. Methodology

This section describes the robust, leakage-controlled experimental pipeline developed for this study, encompassing data preparation, classifier configuration, and the formalization of our multi-axis auditing framework.

A. Dataset

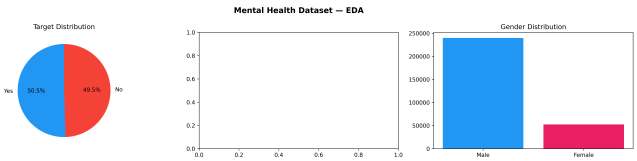


Fig. 1: Target class distribution, age distribution, and self-reported gender distribution in the raw Mental Health Dataset (N = 77,708).

A publicly available dataset [4], comprising 77,708 self-reported survey responses across 17 raw attributes (e.g., gender, country, occupation, self-employment, family history, and several ordinal wellbeing indicators), with a binary target, treatment (has the respondent sought mental-health treatment) is used in this study. After removing a single row with a missing target, the class distribution is 48,125 positive (61.9%) and 29,583 negative (38.1%) instances, as shown in Fig. 1.

B. Leak-Free Preprocessing

Two Non-informative columns, timestamp and free-text comments, are dropped and gender free-text is mapped to a fixed vocabulary (Male/Female/Other) using purely structural rules that involve no dataset statistics. The 80/20 stratified train/test split is performed before any encoder or imputer is fit. A custom `LeakFreePreprocessor` then fit a `LabelEncoder` per categorical column and a per-column median imputer exclusively on the training fold. This ordering guarantees zero information leakage from the test fold into any learned statistic.

C. Class-Imbalance Handling

All but one of the remaining 15 features of the dataset [4] are label-encoded categorical variables. Therefore, standard SMOTE's linear interpolation between neighboring codes is not well defined. We apply SMOTENC [18], which performs

majority-vote resampling on categorical dimensions, exclusively on the training fold (77,000 balanced instances from 62,166) and, independently, inside every cross-validation fold via an imbalanced-learn pipeline, preventing any resampled instance from crossing a fold boundary.

D. Predictive Models

Four classifiers are compared, all seeded at 42 for full reproducibility: XGBoost [19] (200 trees, max depth 5, learning rate 0.05), LightGBM (matched hyperparameters), Random Forest (200 trees, max depth 10), and a scaled Logistic Regression baseline (C = 1.0). Logistic Regression alone requires standardized inputs; its `StandardScaler` is fit exclusively on the resampled training fold.

E. Statistical Model Comparison

Beyond mean $\pm$ SD across 5 stratified CV folds, a Friedman test is run across all four models' per-fold AUC, followed by Holm–Bonferroni-corrected pairwise Wilcoxon signed-rank tests, to determine whether observed differences are statistically defensible given the limited number of folds.

F. Explanation Generation

For tree ensembles, `SHAP TreeExplainer` is used with interventional feature perturbation and the pre-resampling training set as background. We use `SHAP LinearExplainer` on the equivalently scaled background for Logistic Regression. LIME's tabular explainer is fit on the same pre-resampling background. Because Logistic Regression requires scaled inputs while LIME perturbs samples in the raw feature space, we route every LIME per model call through a scaling-aware predict_proba wrapper so that explanation correctness is guaranteed regardless of which model is ultimately selected as best.

G. Four-Axis Explanation Evaluation Framework

The following evaluation framework is implemented in this study:

1) **Faithfulness:** For each instance, features are progressively deleted (by replacing them with their training-set mean) or inserted in descending order of $|\mathrm{SHAP}|$. Hold-out accuracy is recorded over ten perturbation steps. We report the deletion-curve AUC, insertion-curve AUC, their difference (the *faithfulness score*), and the Average Output Probability Change (AOPC) statistic [20].
2) **Stability:** We compute the empirical Lipschitz coefficient [10] for each model's SHAP explanations, defined as the ratio of explanation-space distance to input-space distance over 500 randomly selected pairs of z-score-normalized test instances. For the best-performing model, we additionally measure the variance of repeated LIME explanations for the same instance.
3) **Cross-method agreement:** Agreement between SHAP and LIME for the best-performing model is evaluated using per-instance and global Spearman and Kendall correlations, the top-5 feature-overlap Jaccard index, attribution sign agreement, and global attribution-rank correlation.
4) **Fairness:** Outcome-level fairness is evaluated using demographic-parity, equal-opportunity, and accuracy gaps across the Gender subgroup (the only demographic attribute available in this dataset release) [14]. Explanation-level fairness is assessed using the Explanation Demographic Parity Gap (EDPG), defined for each feature as the difference in mean $|\mathrm{SHAP}|$ values between demographic subgroups. Statistical significance is assessed using the Mann–Whitney U test, with Bonferroni correction applied to account for multiple comparisons [15].

The faithfulness score is reported as insertion-AUC minus deletion-AUC rather than AOPC alone, since a model can degrade quickly under deletion yet recover slowly under insertion (or vice versa) and AOPC would not capture this asymmetry. EDPG is reported as a difference-of-means (not a ratio) to preserve SHAP's native output units and remain directly comparable to the outcome-level parity gap already used in this study.

IV. Results

The following results provide a comparative empirical assessment of model performance followed by a rigorous analysis of explanation reliability and demographic parity across the established evaluation axes.

A. Predictive Performance

TABLE I: Five-Fold Stratified Cross-Validation Performance (Mean $\pm$ SD).

Model	AUC	F1	Accuracy
XGBoost	0.8750 ± 0.0025	0.7874 ± 0.0042	0.7855 ± 0.0044
LightGBM	0.8764 ± 0.0034	0.7881 ± 0.0047	0.7860 ± 0.0048
Random Forest	0.8461 ± 0.0026	0.7628 ± 0.0043	0.7608 ± 0.0044
Logistic Regression	0.7634 ± 0.0036	0.7194 ± 0.0036	0.7168 ± 0.0037

TABLE II: Hold-Out Test-Set Performance.

Model	Accuracy	F1	AUC	Precision	Recall
XGBoost	0.790	0.792	0.876	0.798	0.790
LightGBM	0.784	0.786	0.874	0.792	0.784
Random Forest	0.768	0.770	0.850	0.775	0.768
Logistic Regression	0.719	0.722	0.768	0.729	0.719

Table I summarizes the five-fold cross-validation performance, while the corresponding hold-out test results are presented in Table II. XGBoost achieved the best overall discrimination (CV AUC $= 0.875 \pm 0.003$; hold-out AUC $= 0.876$), followed by LightGBM, Random Forest, and Logistic Regression (hold-out AUCs of 0.874, 0.850, and 0.768, respectively). A Friedman test performed on the per-fold AUCs of the four models revealed a significant overall difference in model rankings ($\chi^2 = 15.00$, $p = 0.0018$), indicating a statistically significant ranking effect. However, none of the six pairwise Wilcoxon signed-rank tests remained significant

after Holm correction ($p_{\text{Holm}} = 0.375$ for all comparisons). This finding is consistent with the limited statistical power of paired nonparametric tests when only five cross-validation folds are available [21]. Therefore, XGBoost was selected as the practical best-performing model because it consistently ranked highest across the evaluation metrics reported in Table II. Nevertheless, the statistical comparison indicates that its advantage over LightGBM is not conclusive, and the two models should be regarded as statistically comparable under the present experimental setting.

B. Faithfulness

TABLE III: Faithfulness Metrics for Each Model.

Model	Del. AUC	Ins. AUC	Faithf.	AOPC
XGBoost	0.528	0.734	0.206	0.222
LightGBM	0.536	0.711	0.175	0.207
Random Forest	0.550	0.720	0.170	0.176
Logistic Regression	0.580	0.670	0.091	0.098

Note: Del. AUC, Ins. AUC, Faithf., and AOP denote deletion AUC, insertion AUC, faithfulness score, and area over the perturbation curve, respectively. For deletion AUC, lower values indicate better faithfulness, whereas higher insertion AUC values indicate better faithfulness.

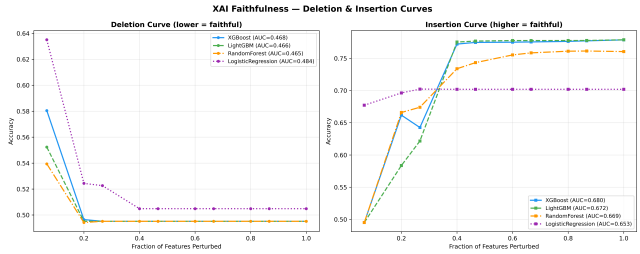


Fig. 2: Deletion (left) and insertion (right) accuracy curves as a function of the fraction of top-ranked SHAP features perturbed, for all four models.

The deletion/insertion faithfulness analysis is summarized in Table III and Fig. 2. XGBoost's SHAP attributions were the most faithful to its own decision function (faithfulness score $= 0.206$, AOPC $= 0.222$), meaning that deleting its top-ranked features degraded hold-out accuracy fastest and inserting them recovered accuracy fastest, compared to the other three models. Logistic Regression explanations were the least faithful (0.091), consistent with its comparatively poor overall fit to this data.

C. Stability

Faithfulness and stability are clearly separated as shown in Table IV and Fig. 4. The SHAP attributions from Random Forest were by far the most Lipschitz–stable (mean $L = 0.051$, 95th percentile $= 0.086$), approximately an order of magnitude more stable than XGBoost (0.388) or LightGBM (0.417), despite ranking third in fidelity. The second best was Logistic Regression (0.231). The boosted trees fit sharper, higher variance decision surfaces that better track their own predictions locally (higher faithfulness), but whose

TABLE IV: Stability of SHAP Attributions (z-Normalized Lipschitz Coefficient).

Explainer	Mean L	95th pct. L
SHAP – XGBoost	0.388	0.701
SHAP – LightGBM	0.417	0.752
SHAP – Random Forest	0.051	0.086
SHAP – Logistic Regression	0.231	0.387
LIME – Best model (var.)	8.8×10^{-4}	1.3×10^{-3}

local gradients and therefore SHAP attributions change more between similar inputs (lower stability). This pattern matches the faithfulness-stability tradeoff described by Yeh et al. [11]. LIME's repeated–explanation variance for the best model (XGBoost) was low in absolute terms (mean $= 8.810^{-4}$) but is not directly comparable in scale to the SHAP Lipschitz statistic.

D. SHAP–LIME Agreement

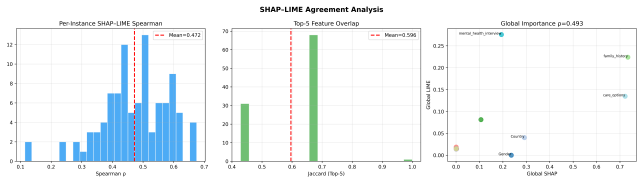


Fig. 3: Per-instance SHAP–LIME Spearman correlation histogram (left), top-5 feature-overlap Jaccard histogram (center), and global feature-importance scatter (right).

TABLE V: SHAP–LIME Agreement Metrics (Best Model: XGBoost, $N = 100$).

Metric	Value
Mean per-instance Spearman ρ	0.488 ± 0.179
Mean per-instance Kendall τ	0.324
Mean top-5 Jaccard overlap	0.590
Mean sign agreement	0.605
Global (aggregate) Spearman ρ	0.679 ($p = 0.0054$)

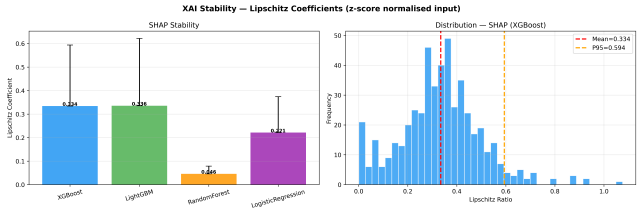


Fig. 4: Mean and 95th-percentile empirical Lipschitz coefficient of SHAP attributions by model (left), and the full distribution of pairwise Lipschitz ratios for the best model (right).

The agreement between the two explanation approaches for the optimal model in Table V and Fig. 3 is of a minor degree. Kendall's τ was 0.324, top-5 feature-overlap Jaccard was 0.590, attribution sign agreement was 0.605 and mean

per-instance Spearman correlation was 0.488 (SD = 0.179). For clinicians who rely on individual-level explanations, take into account that the aggregate (population-level) Spearman correlation between global mean |SHAP| and mean |LIME| was higher at 0.679 ($p = .0054$), meaning that SHAP and LIME agree significantly better on which features matter overall than how to rank features for any given individual.

E. Outcome-Level Fairness

TABLE VI: Outcome-Level Fairness by Gender Subgroup

Subgroup	n	Prev.	TPR	FPR	Acc.	AUC
Female	10,477	0.701	0.893	0.332	0.826	0.880
Male	5,065	0.451	0.459	0.072	0.716	0.827

Note: Prev. = prevalence; TPR = true-positive rate; FPR = false-positive rate; Acc. = accuracy. The demographic-parity gap is 0.479, the equal-opportunity gap is 0.434, and the accuracy gap is 0.110.

The model's results reveal a significant gender imbalance as shown in Table VI. Women in this dataset had a much higher base rate of treatment-seeking (70.1% vs. 45.1% for men), and the model's true-positive rate was correspondingly higher for women (0.893) than men (0.459). This equal-opportunity gap of 0.434 that exceeds what would be expected to arise from the prevalence difference alone, as well as a demographic-parity gap of 0.479 and an accuracy gap of 0.110 in favor of women. It implies that the model is considerably less reliable at correctly identifying men who have sought treatment, a gap that would need to be addressed before any real-world deployment.

F. Explanation-Level Fairness (XAI Audit)

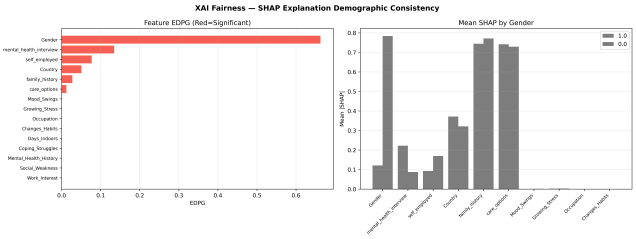


Fig. 5: SHAP explanation-level demographic consistency and feature EDPG by gender subgroup, illustrating significant feature attribution disparities across demographic groups ($p < 0.001$).

The possibility of differences between the explanations based on gender was further examined. In Table VII and Fig. 5, the explanation level demographic parity gap (EDPG) of the difference in the average value of |SHAP| for each feature between genders was presented. Out of 15 features, 14 had a statistically significant EDPG after the Bonferroni correction. These include the Gender feature itself (EDPG = 0.191), Country (0.162), being self-employed (0.135), and care options awareness (0.100), while the one feature which

TABLE VII: Top Explanation-Level Demographic Parity Gaps (EDPG) by Gender (Bonferroni-Corrected).

Feature	EDPG	p (corrected)	Significant?
Gender	0.1910	< 0.001	Yes
Country	0.1620	< 0.001	Yes
Self_employed	0.1350	< 0.001	Yes
Care_options	0.1000	< 0.001	Yes
Mental_health_interview	0.0540	< 0.001	Yes
Family_history	0.0090	< 0.001	Yes
Days_Indoors	0.0007	0.461	No

does not have statistical significance is the Days_Indoors feature ($p = 0.461$). The result means that the reasoning process of our model is gender-biased for almost all features.

G. Global Importance, Interactions, and a Contrastive Case

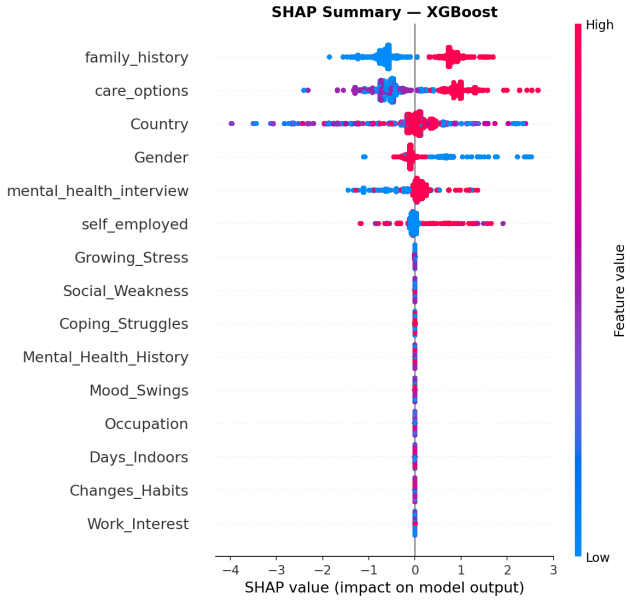


Fig. 6: SHAP beeswarm summary plot of the fifteen features for the best model (XGBoost) on the held-out test set.

The global SHAP summary (Fig. 6) reveals that family_history, care_options, mental_health_interview and self_employed are the strongest predictors based on the same order of priority as the EDPG model ranking in Table VII. The pair–wise SHAP interaction value was minimal in relation to the main effects in all of the top–ranked pairs, implying that XGBoost's learned decision function for this particular problem is near-additive. This might be due to the 100–sample limit imposed on the background for interaction estimations (Section V). Contrastive analysis between one instance that received a treatment-positive prediction and another that received a treatment-negative prediction showed that the most contrasting opposite-signed contribution came from care_options (ΔSHAP = 1.69), family_history (ΔSHAP = 1.67), and Country (ΔSHAP = 0.41).

V. Discussion and Future Work

Our findings suggest that an expert, optimizing only based on AUC value, would unintentionally select the most unstable technique as the model with the highest prediction accuracy. The most faithful explanations (XGBoost) is not the model with most stable explanations (Random Forest), indicating that it is absolutely imperative to provide all three dimensions in conjunction. Another implication is related to the fact that despite the general concordance between techniques with respect to global rankings of features, the choice of explanation technique cannot be interchanged at the instance level given the relatively low SHAP–LIME correlation ($\rho = 0.49$). Third, and perhaps most importantly, there is gender dependency in the explanations and outcomes produced by this model. Either gender stratified re-calibration or training under fairness constraints rather than using one pooled model is required before deploying this sort of screening process, since 14 out of 15 features have a very high EDPG. Finally, the near additive interaction effect observed suggests that an additive model, which would be inherently interpretable, might be able to ensure faithfulness rather than estimate it at the cost of sacrificing little discrimination.

Several constraints need to be taken into account while interpreting these results. Firstly, since the data collection is done through self-reporting from people who do not have any clinical diagnosis, the findings can't be generalized to the population with clinical diagnosis. Secondly, instead of calculating faithfulness based on raw predicted probability which is coarse and often used, here faithfulness is evaluated by hold-out accuracy. In order to make the computation manageable, SHAP interaction values and LIME attributions were computed on sub-samples. The data set lacks the age variable and, therefore, the fairness audit will be limited to gender. The comparison of statistical performance between the models was done using only 5-fold cross-validation since the pair-wise Wilcoxon test lacks enough power despite the significance of the omnibus Friedman test. The label encoding for categorical variables creates an order that SMOTENC does not solve.

This study provides a paradigm of four dimensions (faithfulness, stability, cross-method consistency, and fairness) and a leak-proof and reproducible procedure for auditing SHAP and LIME explanations of table-based mental health treatment prediction models. The results show not only that model predictions and explanations can differ significantly by gender, but also that SHAP and LIME may be individually inconsistent despite demonstrating overall consistency, and that predictive performance, faithfulness, and stability may favor different models. Future research should expand this framework to include intersectional fairness across several demographic axes, validate results on a separate clinically annotated cohort, directly integrate bias-mitigation techniques into the training process, and compare post-hoc explained black-box models with naturally interpretable alternatives under the same four-axis evaluation.

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